

Paper Tables — Full-Sample Two-Grid Run of August 17, 2026

Corrected per the Findings of Record (RUN_FINDINGS_20260817)

Run: replication_20260817_015626 · **Stack:** v0.9.86 · **Sample:** 66 sessions (25 benchmark / 37 volatile / 4 MWCB), 2017-06-26 – 2026-06-05 · **Grids:** 1 s and 10 ms · **Prepared:** August 22, 2026, from the shipped analysis bundle, with the corrections of the Findings of Record applied in place.

Corrections applied (each table's notes give details): (1) Table 5 Woolf z struck as invalid (M-1/M-2); (2) FEVD/GFEVD tables regime-labeled (M-5/E-9); (3) half-life bar-unit fix, $\times 0.01$ at 10 ms (E-3); (4) stale-session tally corrected to 16 benchmark / 9 volatile (I.2); (5) 1 s-only results flagged as such (F-2/F-3); (6) mediation share statistic suppressed (F-3); (7) MWCB taxonomy marked per table (M-8); (8) copula upper-tail LR column omitted pending E-13; (9) Rigobon column excluded, unidentified by its own diagnostic (M-4); (10) pre-averaged θ contradiction flagged unadjudicated (F-4/E-2).

T1. Sample composition and 10 ms estimability

Regime	Sessions	10 ms QC-fail (ES ladder stale)	Share of regime
Benchmark	25	16	64%
Volatile	37	9	24%
MWCB	4	0	0%
All	66	25	38%

QC-fail = ES top of book unchanged on more than 99% of 10 ms snapshots (qc_frames_10ms.txt ok=False); the run estimated on all 66 sessions against this gate (RUN_FINDINGS D-2). The staleness loads on the benchmark side of every regime contrast; the v0.9.87 measured bias curve finds attenuation under the carry-forward mechanism, so 10 ms volatile-minus-benchmark gaps are conservative (D-1). The memo tallies (6/19 and 11/14) were wrong; the correct split is 16 benchmark / 9 volatile / 0 MWCB (I.2). Full per-session roster in Appendix A1.

T2. Table 5 — Crossed-book corner concentration: observed/null ratios and panel log odds ratio

Panel	PCMOF obs/null	NCMOF obs/null	log OR	(Woolf z — invalid)	Corner asymmetry
A — Baseline (benchmark)	1.243	0.763	0.977	(159.8)	-0.016
B — Volatile	1.280	0.730	1.124	(214.3)	-0.019
C — MWCB	1.372	0.659	1.470	(77.7)	-0.019
C — MWCB, ex-SSR member	1.359	0.672	1.413	(64.2)	-0.021

PCMOF/NCMOF: positively/negatively co-moving order-flow corners; ratios are observed corner mass over the independence null. The Woolf z column is retained only for the record and must not be quoted: it is a pooled-bar statistic whose ordering tracks panel size, not effect size (the largest log OR, MWCB 1.470, has nearly the smallest z), and on a planted serially dependent DGP it overstates the day-clustered z by roughly 21x (RUN_FINDINGS M-2); the run-level analogue of the same disease deflates a pooled t of 386.9 to a day-clustered 5.56 (M-1, T15). Day-clustered inference (per-day count matrices, day bootstrap, within-pair sign-flip, MWCB jackknife) ships in v0.9.87 and attaches to the next run. Corner asymmetry approximately -0.02 in every panel: no SSR fingerprint at the pooled level (D-6). MWCB is broken out here; T4-T8 fold or break it out as marked (M-8).

T3. Table 7 — NCMOF excess over the frequency-matched null, by aggregation

Aggregation	Orders per bar (ETF)	Orders per bar (FUT)	NCMOF null %	NCMOF observed %	Observed / null
1 second	505.00	112.00	0.35	11.88	33.54
10 millisecond	5.05	1.12	22.88	30.14	1.32
action time	1.00	1.00	50.08	48.40	0.97

The published aggregation story reproduces on this sample: the negative-corner excess collapses from 33.5x at 1 s bars to 1.32x at 10 ms to 0.97x in action time. No uncertainty ships with this exhibit in this run (RUN_FINDINGS F-6); day-clustered inference attaches with the v0.9.87 machinery on the next run.

T4. Tandem flow coupling (Tier 1) — OFI-innovation correlation by regime and grid

Grid	Regime	Mean corr	Mean z	SE(z)	Days	Permutation p
1 s	Benchmark	0.729	0.926	0.049	25	—
	Volatile	0.818	1.152	0.041	37	—
	MWCB	0.870	1.334	0.039	4	—
	Volatile – benchmark	0.089	0.226	—	62	0.0008
10 ms	Benchmark	0.266	0.273	0.009	25	—
	Volatile	0.292	0.301	0.008	37	—
	MWCB	0.355	0.371	0.008	4	—
	Volatile – benchmark	0.026	0.028	—	62	0.0327

The run's most robust substantive result: coupling orders monotonically with stress on both grids (benchmark < volatile < MWCB), and the volatile-benchmark difference is significant at both ($p = 0.0008$ at 1 s, $p = 0.0327$ at 10 ms). MWCB broken out (4 days). Staleness biases the 10 ms benchmark cell downward (attenuation), making the reported gap conservative (T1 note; RUN_FINDINGS F-1, D-1).

T5. Directional asymmetry of flow coupling — down-move vs up-move bars

Grid	Regime	Down corr (med)	Up corr (med)	Mean dz (down-up)	SE	Sign-flip p	Days
1 s	Benchmark	0.559	0.526	0.0468	0.0154	0.0045	25
	Volatile	0.485	0.563	-0.0901	0.0120	0.0000	37
	MWCB	0.330	0.307	0.0057	0.0154	0.6249	4
	All days	0.514	0.549	-0.0324	0.0121	0.0096	66
10 ms	Benchmark	0.327	0.311	0.0037	0.0033	0.2672	25
	Volatile	0.336	0.335	-0.0023	0.0032	0.4724	37
	MWCB	0.406	0.409	-0.0101	0.0098	0.3727	4
	All days	0.335	0.334	-0.0005	0.0023	0.8224	66

A 1 s result only. At 1 s the benchmark regime couples tighter on down-moves ($+0.047$, $p = 0.0045$) and the volatile regime reverses to tighter on up-moves (-0.090 , $p < 5e-5$). At 10 ms every gap is an order of magnitude smaller and non-significant. The gap-of-gaps is untested in the shipped tables, and the SSR / direction-conditioning checks remain outstanding (RUN_FINDINGS F-2). Quote the reversal as an aggregation-dependent finding or not at all.

T6. Tier 2 mediation — does tandem flow mediate the RV → return-correlation link?

Grid	Regressor	Eq A: Δz_{flow}	Eq B: Δz_{ret} (total)	Eq B: Δz_{ret} (direct)
1 s	rv_es_l1	-5.517** (2.674)	-1.131 (3.300)	5.711** (2.755)
	rv_spy_l1	4.823** (1.943)	0.190 (2.576)	-5.671** (2.388)
	state_l1	-0.094 (0.216)	-0.048 (0.366)	-0.183 (0.296)
	d_wspr_es_l1	0.041 (0.191)	-0.284 (0.269)	0.235 (0.242)
	d_wspr_spy_l1	0.168 (0.224)	0.482 (0.333)	0.416* (0.230)
	d_z_flow (mediator)	--	--	67.009*** (3.766)
	indirect (RV_ES via z_flow)	--	--	-6.842 [-41.940, -3.144]
	indirect share of RV_ES effect	--	--	6.05 [-29.56, 27.36]
10 ms	rv_es_l1	-0.425** (0.201)	-1.076*** (0.350)	-0.790* (0.446)
	rv_spy_l1	1.129*** (0.305)	1.831*** (0.442)	0.890* (0.466)
	state_l1	0.052 (0.060)	0.115 (0.087)	0.014 (0.056)
	d_wspr_es_l1	0.044 (0.060)	-0.024 (0.077)	0.036 (0.064)
	d_wspr_spy_l1	0.064 (0.081)	0.206* (0.110)	0.126** (0.051)
	d_z_flow (mediator)	--	--	75.270*** (1.183)
	indirect (RV_ES via z_flow)	--	--	-0.286 [-2.346, 0.163]
	indirect share of RV_ES effect	--	--	0.27 [-0.14, 1.21]

Day-cluster bootstrap SEs in parentheses; */**/** = 10/5/1%. The mediator link ($\Delta z_{\text{flow}} \rightarrow \Delta z_{\text{ret}}$: 67.0 at 1 s, 75.3 at 10 ms) is the strongest coefficient in the run and holds on both grids. The suppression decomposition is a 1 s result: there the indirect level CI [-41.9, -3.14] excludes zero; the share row (6.05 = 605%) is a ratio to a near-zero total and must not be quoted. At 10 ms the indirect CI includes zero. Bootstrap resolution is unequal across grids (499 vs 199 draws; RUN_FINDINGS F-3, D-7); the next-run config re-runs both at $n_{\text{boot}} = 2000$ with BCa intervals.

T7. Lead-lag at 10 ms — HRY timing estimate and co-jump ordering

Test	Point estimate	Days ES-leading	Day-level sign-flip p
HRy cross-correlation θ (ES leads by)	21.5 ms (median 20.0 ms)	64 / 66	0.0005
Co-jump lead share (ES first)	0.358	65 / 66	5.0e-05

Both tests are day-level (immune to the pooled-inference disease of M-1). ES leads SPY by about one 20 ms grid step. Integrity notes from RUN_FINDINGS F-4: (i) the co-jump ES-lead share correlates POSITIVELY with ES update counts (Spearman +0.57) — weakest on stale sessions, so not a staleness artifact; (ii) the two extreme raw- θ days (2017-06-26, 2017-10-05) are the two stalest sessions and should be excluded from θ magnitude claims; (iii) the pre-averaged θ column reverses sign on 62/66 days and ships untested — the raw-vs-pre-averaged contradiction is UNADJUDICATED on this record and the ES-lead claim should cite the co-jump test first. At 1 s neither test resolves ($p = 0.52/0.96$): a resolution artifact.

T8. Information-share regime tests — volatile vs benchmark

Grid	Test	Volatile mean	Benchmark mean	Difference	p
1 s	CS_ES, free permutation	0.336	0.302	0.034	0.5491
	IS_mid_ES, free permutation	0.469	0.437	0.032	0.0324
	CS_ES, within-pair (era-robust)	—	—	-0.002	0.9755
	CS_ES, κ -weighted	kappa	0.292	0.263	0.0285
10 ms	CS_ES, free permutation	0.398	0.355	0.043	0.1169
	IS_mid_ES, free permutation	0.484	0.437	0.047	0.1253
	CS_ES, within-pair (era-robust)	—	—	0.038	0.1324
	CS_ES, κ -weighted	kappa	0.354	0.343	0.0109

Within-pair rows report the mean pair difference over 25 pairs. The only sub-0.05 cell is the 1 s free-permutation IS test ($p = 0.032$) — and its era-robust within-pair counterpart DOES NOT EXIST in this run (it was run only on CS, where it reverses sign at 1 s, $p = 0.976$). The claim "ES gains share in volatile regimes" is therefore unsupported by an era-robust test on this record (RUN_FINDINGS F-5). v0.9.87 adds the within-pair and ex-straddle twins on IS_mid_ES; the next run carries the verdict. IS is the primary metric; CS is fragile to lags and beta (M-7). Volatile here folds MWCB in (M-8).

T9. Table 9 — IRF of return-correlation change on liquidity/flow shocks (1 s), Pearson vs HY vs DCC

Shock	Pearson bench	Pearson vol	HY bench	HY vol	DCC bench	DCC vol	$\Delta(\text{HY-P})$ bench	$\Delta(\text{HY-P})$ vol
Spread_ES	0.003 (0.001)	0.000 (0.001)	0.006** (0.001)	0.001 (0.001)	0.000 (0.001)	-0.000 (0.001)	0.003	0.001
WtdSpread_ES	0.001 (0.005)	-0.000 (0.002)	-0.048** (0.014)	-0.011** (0.003)	0.021*** (0.005)	0.011*** (0.002)	-0.049	-0.011
OFI_ES	-0.015 (0.012)	-0.000 (0.003)	0.018 (0.014)	-0.012 (0.005)	-0.005 (0.004)	-0.001 (0.002)	0.033	-0.012
RV_ES	0.178 (0.061)	0.041* (0.012)	0.137 (0.046)	0.028 (0.010)	-0.004 (0.003)	0.001 (0.002)	-0.041	-0.013
MicroDev_ES	0.003 (0.005)	-0.003 (0.001)	-0.000 (0.007)	-0.004 (0.001)	0.001 (0.002)	-0.001 (0.001)	-0.003	-0.002
Spread_SPY	0.004 (0.002)	0.000 (0.001)	0.002 (0.002)	0.000 (0.001)	0.002 (0.002)	-0.000 (0.001)	-0.002	-0.000
WtdSpread_SPY	0.023** (0.005)	0.014** (0.003)	0.039** (0.010)	0.013 (0.004)	-0.004 (0.003)	0.001 (0.001)	0.016	-0.001
OFI_SPY	0.004 (0.004)	0.004 (0.002)	0.010 (0.007)	-0.000 (0.002)	-0.005 (0.002)	0.002 (0.001)	0.006	-0.004
RV_SPY	-0.085 (0.034)	-0.004 (0.015)	-0.152* (0.045)	0.002 (0.016)	-0.005 (0.006)	0.001 (0.001)	-0.068	0.005
MicroDev_SPY	-0.004 (0.003)	0.001 (0.001)	0.002 (0.003)	0.001 (0.001)	0.000 (0.002)	-0.000 (0.001)	0.007	-0.000

Impact orthogonalized responses of $\Delta\text{correlation}$, $\times 100$; panel VAR(5) with day fixed effects, within-day lags; day-cluster bootstrap SEs, Romano-Wolf joint stars (***/**/* = 1/5/10% FWER). The Δ column is the share of each published Pearson response attributable to Epps-type measurement artifact. Standing caveats (RUN_FINDINGS): the 10 ms leg of this table died at ~292.6 GiB and does not exist (I.1); no lag-band certification ran, so no cell is certified lag-robust (I.3); DCC is a boundary estimate ($a+b = 0.9999$, M-6) — read its near-zero responses accordingly; the RealBar column is omitted here pending the lag diagnostic (available in the bundle). The v0.9.87 local-projection twin (T9_LP=1) replaces lag-order-dependent IRFs next run (M-3).

T10. Forecast-error variance of SPY returns attributed to ES order flow — diagonal FEVD vs generalized (GFEVD), by regime (1 s)

Decomposition	Regime	SPY ret $\leftarrow$ OFI_SPY	SPY ret $\leftarrow$ OFI_ES
FEVD (diagonal — invalid here)	Benchmark	0.721	0.279
	Volatile	0.495	0.505
GFEVD (Pesaran–Shin — quote this)	Benchmark	0.543	0.457
	Volatile	0.500	0.500

LABELING CORRECTION applied: the shipped `irf__fevd.csv` / `irf__gfevd.csv` are the BENCHMARK-regime tables with no regime label; the volatile decompositions existed only inline in `report.md` (RUN_FINDINGS M-5, E-9). Measured OFI-innovation correlation is 0.705 (benchmark) / 0.788 (volatile), so the diagonal FEVD's orthogonality assumption fails and the GFEVD is the quotable object — with the caveat that under tandem flow, common flow is counted toward both shocks. Roughly 46-50% of SPY-return forecast-error variance attributes to ES flow.

T11a. Copula tail dependence by regime (1 s, 15 m windows)

Regime	Best family (days)	Median λ_L	Median λ_U	Tail asym (L–U)	Sign-flip p	Days
Benchmark	t:19 sjc:4 bb1:2	0.531	0.531	0.0197	0.2490	25
Volatile	sjc:20 bb1:11 t:6	0.685	0.688	-0.0023	0.6866	37
MWCB	t:2 frank:2	0.389	0.389	-0.0287	0.1253	4
All days	t:27 sjc:24 bb1:13 frank:2	0.622	0.628	0.0044	0.5475	66

Parametric caveat (RUN_FINDINGS): the t family FORCES $\lambda_L = \lambda_U$, and t wins 19/25 benchmark days — the benchmark row's symmetry is partly assumed, not measured. The v0.9.87 nonparametric Schmidt-Statdmüller estimator (λ at $q = 0.95$, no family constraint) attaches next run and adjudicates. The upper-tail LR column (median $p = 0.0$ in all groups) is omitted pending the E-13 column-definition fix.

T11b. Lower-tail dependence conditional on book state — thin vs deep liquidity (1 s)

Regime	Median λ_L thin	Median λ_L deep	Δ (thin–deep)	SE	Sign-flip p	Days
Benchmark	0.503	0.448	-0.0203	0.0402	0.6111	25
Volatile	0.641	0.551	0.0762	0.0128	0.0000	37
MWCB	0.813	0.812	0.0196	0.0240	0.7461	4
All days	0.597	0.506	0.0362	0.0176	0.0443	66

In the volatile regime, tail dependence is significantly higher when the book is thin (+0.076, $p = 5e-5$) — the liquidity channel of tail coupling. Benchmark and MWCB do not resolve.

T12. Markov-switching flow-coupling regimes within the day (1 s)

Regime	Days by chosen K	Corr levels (modal K)	Top-regime share	Median duration lo/hi (bars)	Duration asym p	Entry ret (bps)	Entry-direction p	Multi-regime days
Benchmark	K=1:1 K=2:10 K=3:13 K=4:1	0.64 / 0.76 / 0.89	0.23	156 / 36	0.0007	-0.31	0.587	24 of 25
Volatile	K=1:1 K=2:22 K=3:14	0.80 / 0.87	0.39	81 / 61	0.4151	-1.60	0.221	36 of 37
MWCB	K=2:3 K=4:1	0.67 / 0.89	0.73	9 / 197	0.1263	-22.08	0.125	4 of 4
All days	K=1:2 K=2:35 K=3:27 K=4:2	0.75 / 0.87	0.35	89 / 54	0.0769	-2.76	0.020	64 of 66

64 of 66 days reject a single flow-coupling regime (median LR $p = 0.01$ — a 99-draw resolution FLOOR, not an estimate; RUN_FINDINGS D-7). Benchmark days spend long spells in the low-coupling state punctuated by short high-coupling bursts (156 vs 36 bars, asymmetry $p = 0.0007$); volatile days are more evenly split. Regime entries lean toward down-moves overall (entry-direction $p = 0.020$, all days).

T13. Horizon profile — cross-asset transmission across sampling rungs (66 days)

Interval	λ SPY $\leftarrow$ ES	λ ES $\leftarrow$ SPY	IS_mid_ES	CS_ES	Half-life (bars)	Half-life (s, corrected)	ES lead share
10ms	0.038	0.015	0.466	0.381	5.4	0.05	0.358
50ms	0.152	0.087	0.413	0.329	26.2	1.31	0.086
100ms	0.180	0.007	0.413	0.311	67.9	6.79	0.041
250ms	0.250	0.013	0.429	0.310	56.2	14.06	-0.039
500ms	0.309	0.021	0.444	0.317	57.4	28.69	-0.069
1s	0.370	0.033	0.457	0.323	63.9	63.92	0.006

UNIT CORRECTION applied: the shipped half_life_s column is in BAR units; true seconds = bars x bar length (RUN_FINDINGS E-3 — a $\times 100$ label error at 10 ms). Corrected, the error-correction half-life runs 0.05 s at 10 ms to 64 s at 1 s bars. The asymmetric cross-impact (λ SPY $\leftarrow$ ES an order of magnitude above λ ES $\leftarrow$ SPY at every rung) and the ES lead share visible only below 100 ms are the profile's substantive content.

T14. ECM-SDE error-correction estimates (Kalman, curve-implied)

Grid	a1 SPY (t)	a1 ES (t)	Median IS_ES (mid)	Median IS_ES (microprice)	Median half-life (s, corrected)
1 s	0.00067 (0.30)	-0.00972 (-3.33)	0.176	0.308	9.4
10 ms	0.00022 (3.68)	-0.00024 (-6.07)	0.569	0.646	1.45

Signs are the textbook error-correction pattern (SPY adjusts up toward, ES down toward, the common curve). Half-life carries the same bar-unit correction as T13 ($\times 0.01$ at 10 ms; RUN_FINDINGS E-3). Microprice-based IS is above mid-based on both grids. t-statistics are as shipped (per-day Kalman aggregation).

T15. Inference exhibit — pooled-iid vs day-clustered t on the same regression (1 s)

Regression	Coefficient	Pooled-iid t	Day-clustered t	Wild-bootstrap p	N obs	N days
SPY_ret ~ ES_OFI	1.05e-08	386.9	5.56	0.001	1529994	66

The 69.5x deflation that governs the reading of every pooled statistic in the run (RUN_FINDINGS M-1), and the reason T2's Woolf z column is struck. All starred results quoted in this document use day-level or day-clustered inference.

T16. Cross-impact regime summary (10 ms)

Regime	Type	Mean	SD	Day-windows
Benchmark	cross	0.0118	0.0215	50
Benchmark	own	0.8417	0.0442	50
Volatile (incl. MWCB)	cross	0.0352	0.0412	82
Volatile (incl. MWCB)	own	0.7940	0.0996	82

Cross-impact triples from benchmark (0.012) to volatile (0.035) while own-impact barely moves. Taxonomy note: this family folds MWCB into volatile (41 days; M-8).

A1. Appendix — per-session 10 ms QC roster

Date	Regime	ES top-of-book stale fraction	10 ms QC	Notes
2017-06-26	bench	0.9986	fail	ladder-monotonicity defect (8 snaps); session0 exemplar
2017-08-10	vol	0.9970	fail	
2017-09-05	vol	0.9967	fail	
2017-10-05	bench	0.9994	fail	
2017-10-25	bench	0.9980	fail	
2017-11-28	bench	0.9985	fail	
2018-01-29	vol	0.9972	fail	
2018-02-02	vol	0.9924	fail	
2018-02-05	vol	0.9763	ok	
2018-03-19	vol	0.9910	fail	
2018-03-22	vol	0.9847	ok	
2018-03-23	bench	0.9810	ok	
2018-05-29	vol	0.9918	fail	
2018-06-25	vol	0.9924	fail	
2018-08-06	bench	0.9986	fail	
2018-10-04	vol	0.9942	fail	

2018-10-10	vol	0.9860	ok	
2018-12-04	vol	0.9799	ok	
2019-03-22	vol	0.9900	ok	
2019-05-14	bench	0.9940	fail	
2019-06-13	bench	0.9958	fail	
2019-08-05	vol	0.9866	ok	
2019-09-05	bench	0.9954	fail	
2019-10-28	bench	0.9977	fail	
2020-01-27	vol	0.9911	fail	
2020-01-29	bench	0.9938	fail	
2020-02-24	vol	0.9832	ok	
2020-02-27	bench	0.9531	ok	
2020-03-09	MWCB	0.9275	ok	MWCB halt rows masked
2020-03-12	MWCB	0.8876	ok	MWCB halt rows masked
2020-03-16	MWCB	0.9012	ok	MWCB halt rows masked; SSR 100%
2020-03-18	MWCB	0.8852	ok	MWCB halt rows masked
2020-05-12	vol	0.9862	ok	
2020-06-11	vol	0.9657	ok	
2020-09-03	vol	0.9442	ok	
2020-10-26	vol	0.9765	ok	
2020-12-11	bench	0.9853	ok	activity rule extracted ESZ0 (75.8%); QC cols score calendar front (D-5)
2021-01-27	vol	0.9598	ok	
2021-02-04	bench	0.9916	fail	

2021-02-25	vol	0.9478	ok	
2021-11-26	vol	0.9510	ok	early close 13:00; post-close rows masked
2022-02-10	vol	0.9426	ok	SSR 100%
2022-06-13	vol	0.9418	ok	
2023-03-09	vol	0.9762	ok	
2023-07-26	bench	0.9850	ok	
2023-08-07	bench	0.9904	fail	
2023-09-05	bench	0.9915	fail	
2023-12-20	bench	0.9904	fail	
2024-01-29	bench	0.9939	fail	
2024-02-23	bench	0.9893	ok	
2024-04-04	bench	0.9854	ok	
2024-07-24	vol	0.9829	ok	
2024-08-02	bench	0.9287	ok	
2024-08-05	vol	0.8874	ok	
2024-09-03	vol	0.9726	ok	
2024-10-11	bench	0.9908	fail	
2024-12-18	vol	0.9651	ok	
2025-01-21	bench	0.9805	ok	
2025-01-27	vol	0.9613	ok	
2025-02-21	vol	0.9751	ok	
2025-04-03	vol	0.9091	ok	
2025-06-06	bench	0.9785	ok	

2025-08-01	vol	0.9573	ok
2025-10-10	vol	0.9082	ok
2026-01-20	vol	0.9547	ok
2026-06-05	vol	0.9279	ok

QC fail threshold: ES top of book unchanged on >99% of snapshots. The 1 s frames pass QC on all 66 sessions; every 10 ms exhibit in this document estimated on all 66 (D-2) — the clean-subsample and dose-response (es_stale_frac) checks attach next run.